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Course: Calculus Beta

Assignment: MyLab Math - Uge 13

Find ∇f at the given point.

$$f(x,y,z) = (x^2 + y^2 + z^2)^{-1/2} + \ln(xyz), \quad (-2, -1, 2)$$

The gradient of $f(x,y,z)$ at any given point is the vector $\frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}$, obtained by taking the partial derivatives at that point.

Start by finding $\frac{\partial f}{\partial x}$, using the chain rule and simplifying the fractions. Recall that $\frac{d(\ln x)}{dx} = \frac{1}{x}$.

$$\begin{aligned} \frac{\partial f}{\partial x} &= (2x) \left(-\frac{1}{2} \right) (x^2 + y^2 + z^2)^{-3/2} + \frac{yz}{xyz} \\ &= -\frac{x}{(x^2 + y^2 + z^2)^{3/2}} + \frac{1}{x} \end{aligned}$$

Use the same procedure to find $\frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial z}$.

$$\begin{aligned} \frac{\partial f}{\partial y} &= -\frac{y}{(x^2 + y^2 + z^2)^{3/2}} + \frac{1}{y} \\ \frac{\partial f}{\partial z} &= -\frac{z}{(x^2 + y^2 + z^2)^{3/2}} + \frac{1}{z} \end{aligned}$$

Evaluate $\frac{\partial f}{\partial x}$ at $(-2, -1, 2)$.

$$\begin{aligned} \frac{\partial f}{\partial x} &= -\frac{x}{(x^2 + y^2 + z^2)^{3/2}} + \frac{1}{x} \\ \left(\frac{\partial f}{\partial x} \right)_{(-2, -1, 2)} &= -\frac{-2}{((-2)^2 + (-1)^2 + (2)^2)^{3/2}} + \frac{1}{-2} \\ &= -\frac{23}{54} \end{aligned}$$

Evaluate $\frac{\partial f}{\partial y}$ at $(-2, -1, 2)$.

$$\begin{aligned} \frac{\partial f}{\partial y} &= -\frac{y}{(x^2 + y^2 + z^2)^{3/2}} + \frac{1}{y} \\ \left(\frac{\partial f}{\partial y} \right)_{(-2, -1, 2)} &= -\frac{(-1)}{((-2)^2 + (-1)^2 + (2)^2)^{3/2}} + \frac{1}{(-1)} \\ &= -\frac{26}{27} \end{aligned}$$

Evaluate $\frac{\partial f}{\partial z}$ at $(-2, -1, 2)$.

$$\begin{aligned} \frac{\partial f}{\partial z} &= -\frac{z}{(x^2 + y^2 + z^2)^{3/2}} + \frac{1}{z} \\ \left(\frac{\partial f}{\partial z} \right)_{(-2, -1, 2)} &= -\frac{2}{((-2)^2 + (-1)^2 + (2)^2)^{3/2}} + \frac{1}{2} \\ &= \frac{23}{54} \end{aligned}$$

Finally, use $\left(\frac{\partial f}{\partial x} \right)_{(-2, -1, 2)} = -\frac{23}{54}$, $\left(\frac{\partial f}{\partial y} \right)_{(-2, -1, 2)} = -\frac{26}{27}$, and $\left(\frac{\partial f}{\partial z} \right)_{(-2, -1, 2)} = \frac{23}{54}$ to find the gradient. Each partial derivative corresponds to one of the components of $\nabla f|_{(-2, -1, 2)}$.

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$$

$$\nabla f|_{(-2, -1, 2)} = -\frac{23}{54} \mathbf{i} - \frac{26}{27} \mathbf{j} + \frac{23}{54} \mathbf{k}$$